LABORATORY OBSERVATION

Course: Full Stack Web Development
Course Code: 23CM4121
Branch: CSM
Section: B
Task No.: 1
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1. TITLE

Design and Develop a College Information Portal Using HTML and HTML5 to Demonstrate Semantic Structure, Text Formatting, Lists, Tables, Forms, and Multimedia Elements.

2. AIM

To design and develop a static HTML5 web page — the "College Information Portal" — that demonstrates the use of semantic tags, text formatting elements, ordered and unordered lists, hyperlinks, a structured table, images, an interactive registration form, HTML5 input types, and embedded audio/video elements.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic structure and syntax of an HTML5 document.
- To use semantic tags such as header, nav, section, article, and footer to organise page content.
- To apply text formatting tags such as b, i, u, sup, and sub.
- To create ordered and unordered lists to present information.
- To insert hyperlinks that open external websites in a new tab.
- To design a table to display structured student data.
- To embed images with meaningful alt text.
- To build an interactive registration form using text, email, password, radio, checkbox, select, and textarea inputs.
- To use HTML5 input types such as date, number, color, and range.
- To embed audio and video elements using native HTML5 support.

4. THEORY

HTML (HyperText Markup Language) is the standard markup language used to create the structure of web pages. HTML5 is the latest version of HTML. It introduces semantic elements and new form input types that make pages more meaningful and interactive, without needing external plugins or JavaScript.

In this page, semantic and formatting tags are used to structure content and present it clearly to the browser.

- header, nav, section, article, footer are used to organise related content into logical blocks.
- sup and sub raise or lower text — used here for ordinal suffixes and small notes.

- ol and ul create ordered and unordered lists respectively.
- a href="..." target="_blank" creates a hyperlink that opens in a new tab.
- table, tr, th, td define a table, its rows, header cells, and data cells.
- form, input, select, textarea create interactive elements for user input.
- audio and video embed native media players without external plugins.

The basic flow of the page is:

Browser requests the HTML file → Browser parses the HTML → Elements are rendered as a structured web page

5. ALGORITHM / PROCEDURE

1. Create a new HTML file named college_portal.html.
2. Add the DOCTYPE declaration and set the language attribute.
3. Add the head section with meta tags and a page title.
4. Add a header section containing the college name, followed by a navigation bar with Home, About, and Contact links.
5. Create a section with formatted text, an ordered list of courses, and an unordered list of facilities.
6. Insert hyperlinks to external websites, set to open in a new tab.
7. Create a table to display student roll number, name, branch, and email.
8. Insert images for the college logo and campus.
9. Build a registration form using text, email, password, radio, checkbox, select, and textarea inputs.
10. Add an article block showing a notice.
11. Embed audio and video elements using the audio and video tags.
12. Add HTML5 input types such as date, number, color, and range.
13. Add a footer with copyright information.
14. Save the file and open it in a web browser.
15. Verify that all elements render and function correctly.

6. PROGRAM

college_portal.html

```
<!DOCTYPE html>
<!-- Declares this document as HTML5 -->
<html lang="en">
<head>
<meta charset="UTF-8">
<title>College Information Portal</title>
</head>
<body>

<!-- ================ PART A: BASIC HTML, TEXT FORMATTING & LISTS ================ -->
<header>
  <!-- <header> marks the introductory banner of the page -->
  <h1>ANITS Engineering College</h1>
</header>

<nav>
  <!-- <nav> groups the site's navigation links together -->
```

```html
    <a href="#">Home</a> | <a href="#">About</a> | <a href="#">Contact</a>
  </nav>

  <section>
    <p>Welcome to our college portal. We offer <b>quality education</b>,
    <!-- <b> bolds a keyword for emphasis -->
       <i>modern infrastructure</i>, and <u>excellent placements</u>.</p>
       <!-- <i> italicises, <u> underlines - simple visual formatting -->

    <p>Established in 2005 | Our college secured 1<sup>st</sup> rank
    <!-- <sup> raises "st" above the line, same as it is written by hand -->
       in the state for placements<sub> (2024)</sub>.</p>
       <!-- <sub> lowers the year below the line, used here as a small note -->

    <h2>Courses Offered</h2>
    <!-- Ordered list: the courses have a fixed order of listing -->
    <ol>
      <li>CSE</li>
      <li>ECE</li>
      <li>MECH</li>
    </ol>

    <h2>Facilities</h2>
    <!-- Unordered list: no ranking implied between facilities -->
    <ul>
      <li>Library</li>
      <li>Sports Complex</li>
      <li>Hostel</li>
    </ul>

     <p><a href="https://www.college-example.edu" target="_blank">College Website</a> |
    <!-- target="_blank" opens the link in a new browser tab -->
       <a href="https://www.google.com" target="_blank">Google</a></p>
  </section>

  <!-- ================= PART B: TABLE & IMAGES ================= -->
  <section>
    <h2>Student Details</h2>
    <table border="1" cellpadding="8">
      <!-- <tr> = table row, <th> = header cell, <td> = data cell -->
      <tr><th>Roll No</th><th>Name</th><th>Branch</th><th>Email</th></tr>
        <tr><td>101</td><td>Saranya</td><td>CSE</td><td>saranya@example.com</td></tr>
      <tr><td>102</td><td>Ravi</td><td>ECE</td><td>ravi@example.com</td></tr>
    </table>

    <h2>Logo &amp; Campus</h2>
    <!-- alt text describes the image if it fails to load -->
    <img src="logo.png" alt="ANITS Engineering College Logo" width="120">
    <img src="campus.jpg" alt="ANITS Engineering College Campus View" width="300">
  </section>

  <!-- ================= PART C: STUDENT REGISTRATION FORM ================= -->
  <section>
    <h2>Student Registration Form</h2>
    <form>
      Name: <input type="text" required><br>
      <!-- "required" is HTML5 built-in validation, no JavaScript needed -->
```

```html
    Email: <input type="email" required><br>
    <!-- type="email" checks for a valid email pattern before submit -->
    Password: <input type="password" required><br>
    <!-- type="password" masks the typed characters -->
    Gender: <input type="radio" name="gender">Male
            <input type="radio" name="gender">Female<br>
    <!-- shared name="gender" links both radios into one group -->
    Hobbies: <input type="checkbox">Reading
             <input type="checkbox">Sports<br>
    <!-- checkboxes act independently, so multiple can be ticked -->
    Branch:
    <select>
      <option>CSE</option>
      <option>ECE</option>
      <option>MECH</option>
    </select><br>
    Comments:<br>
    <textarea rows="4" cols="30"></textarea><br>
    <!-- <textarea> allows multi-line input, unlike a single-line <input> -->
    <input type="submit"><input type="reset">
  </form>
</section>

<!-- ================= PART D: SEMANTIC BLOCK, HTML5 INPUTS & MEDIA =================
-->
<section>
  <article>
    <!-- <article> is content that makes sense as its own standalone block -->
    <h3>Latest Notice</h3>
    <p>Semester exams start from next month.</p>
  </article>

  <h2>Media</h2>
  <audio controls src="welcome.mp3"></audio>
  <!-- controls = shows the built-in play/pause/volume bar automatically -->
  <video controls width="300" src="campus_tour.mp4"></video>

  <h2>HTML5 Input Types</h2>
  <form>
    Email: <input type="email"><br>
    DOB: <input type="date"><br>
    <!-- type="date" opens a calendar date-picker -->
    Age: <input type="number" min="17" max="30"><br>
    <!-- type="number" restricts entry to numbers within a range -->
    Color: <input type="color"><br>
    <!-- type="color" opens a colour-picker swatch -->
    Rating: <input type="range" min="1" max="5">
    <!-- type="range" gives a slider instead of typing a number -->
  </form>
</section>

<footer>
  <!-- <footer> marks the closing section of the page -->
  <p>&copy; 2026 ANITS Engineering College</p>
</footer>

</body>
```

```
</html>
```

7. CODE EXPLANATION

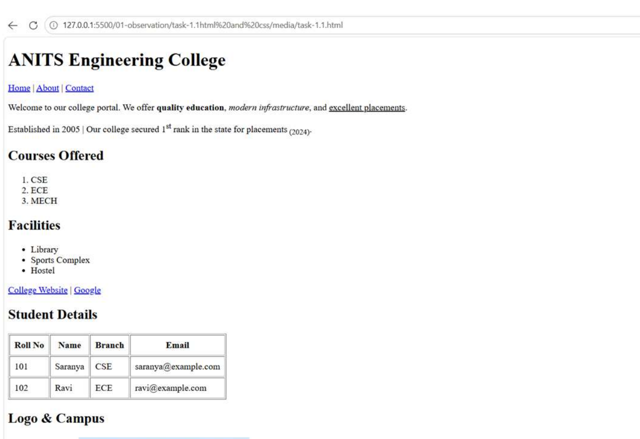
Tag / Attribute	Description
<header>, <nav>, <footer>	Group the banner, navigation links, and closing section into semantic, meaningful blocks.
<section>, <article>	Group related content and mark out a standalone block, respectively.
<sup>, <sub>	Raise or lower text for ordinal suffixes and small notes.
<b>, <i>, <u>	Bold, italicise, and underline text for visual emphasis.
, ,	Create ordered and unordered lists and their items.
<a href="..." target="_blank">	Creates a hyperlink that opens in a new browser tab.
<table>, <tr>, <th>, <td>	Define a table, its rows, header cells, and data cells.
<img src="" alt="">	Embeds an image with alternate text for accessibility.
<form>, <input>, <select>, <textarea>	Create the interactive elements of the registration form.
required	HTML5 built-in validation that blocks submission if a field is empty.
type="radio" / "checkbox"	Radio buttons allow one choice per group; checkboxes allow multiple.
type="date" / "number" / "color" / "range"	HTML5 input types for specialised data entry.
<audio controls>, <video controls>	Embed native audio/video players without external plugins.

8. TEST CASES AND EXECUTION DATA

The page was opened in a browser and each interactive element was tested individually; the results were recorded below.

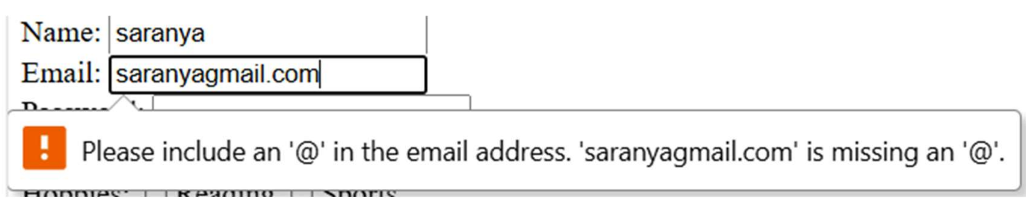
Test Case: TC-01
Action: Open college_portal.html in a web browser
Expected Result: The header, navigation bar, semantic sections, structured student table, images, registration form, and footer should render correctly.
Actual Result: All HTML5 structural and text formatting elements loaded properly in initial state.
Status: Pass

TC-01 Output:

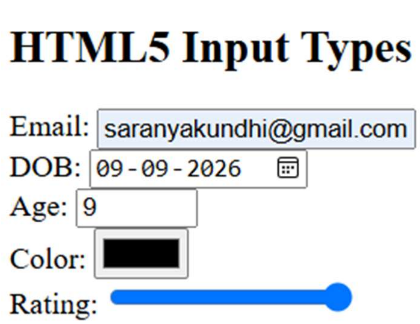


Test Case: TC-02
Action: Click the 'Submit' button on the Student Registration Form without entering any data
Expected Result: Browser blocks form submission and displays a native HTML5 required-field validation prompt on the Name input field.
Actual Result: Submission was prevented and "Please fill out this field" validation tooltip appeared.
Status: Pass

Test Case: TC-03
Action: Enter an invalid email format (e.g., saranya.com without @) into the Email field and click Submit
Expected Result: Browser flags the email pattern error and blocks submission.
Actual Result: HTML5 email validation caught the missing @ symbol and displayed an error tooltip.
Status: Pass

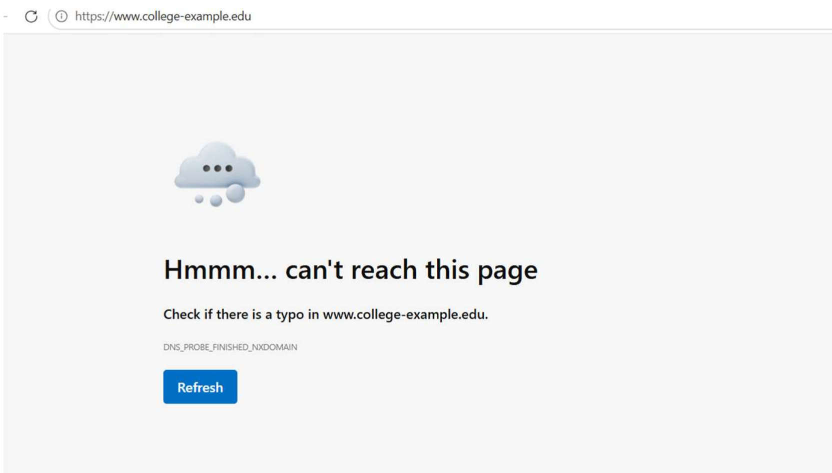


Test Case: TC-04
Action: Enter an Age value outside the specified range (e.g., enter 15 or 35 when min="17" and max="30")
Expected Result: HTML5 number input validation blocks form submission due to out-of-bounds range.
Actual Result: Browser showed validation error stating value must be greater than or equal to 17 / less than or equal to 30.
Status: Fail (Validation Triggered)

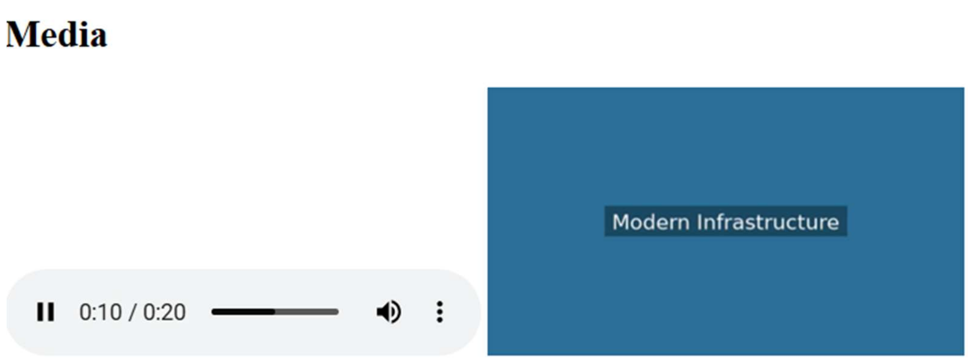


Test Case: TC-05
Action: Click the 'College Website' external hyperlink (target="_blank")

Expected Result: The external link opens in a new browser tab without replacing the current portal page.
Actual Result: Destination URL opened successfully in a new tab.
Status: Pass



Test Case: TC-06
Action: Click the Play button on embedded <audio> and <video> media controls
Expected Result: Native browser media player streams welcome.mp3 and campus_tour.mp4 smoothly.
Actual Result: Audio track and video player executed playback with functional controls.
Status: Pass

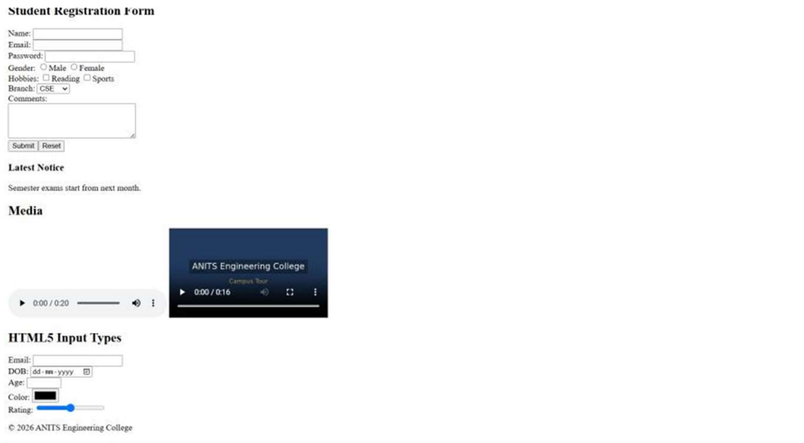


9. OUTPUT

Output 1: Basic HTML

Output 2: Table & Images

Output 3: HTML Forms and Media



10. RESULT

Thus, a College Information Portal was successfully designed and developed using HTML5. Semantic tags, text formatting, ordered and unordered lists, hyperlinks, a structured table, images, a registration form, HTML5 input types, and multimedia elements were implemented and verified to render and function correctly in the web browser.